

The Architect's Guide to Prosthodontics: Biomechanics & Immediate Rehabilitation

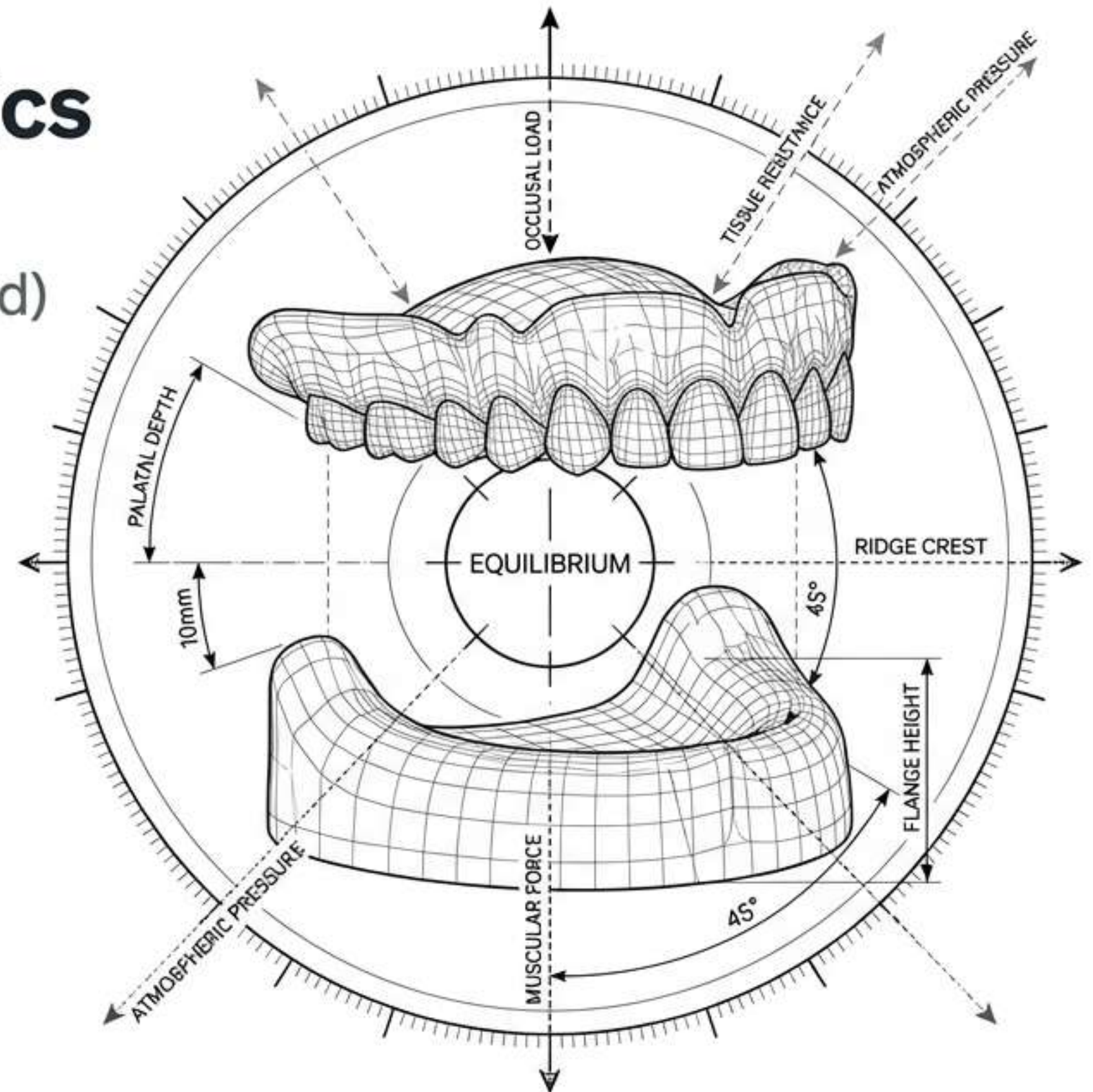
Complete Denture Physics (The Housset Triad)
& The Surgical-Prosthetic Workflow of Immediate Complete Dentures (PCIU)

Master Table of Contents

Part I: The Blueprint of Equilibrium
(Retention, Stability, Sustentation)

Part II: The Engineering of Immediate
Dentures (PCIU)

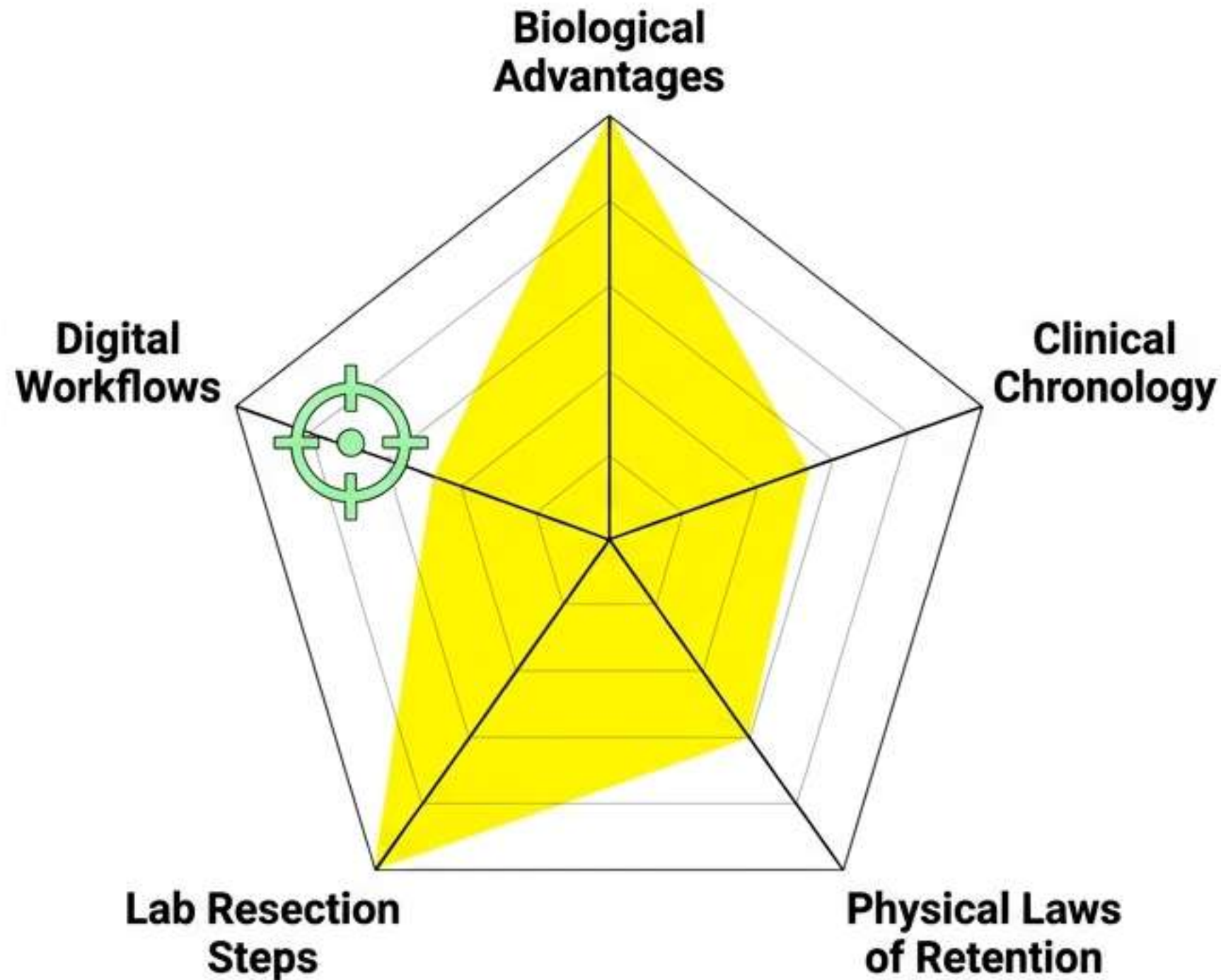
Part III: Clinical & Laboratory Workflows



Zero-Omission Study Protocol Active. Supreme Authority: PDF Texts.

Exam Pattern Analysis & Intelligence Strategy

Radar Chart



Most Repeated Topics

- **Biological advantages of PCIU** (Pansement chirurgical, bone preservation)
- **Chronological lab steps** (Plaster resection)

Frequently Tested Traps

- **The exact timing of extractions** (Posterior first)
- **The biological effect on bone** (Decreases resorption, does not accelerate it)

Critical Authority Conflict (Supreme PDF Rule)

Exam Q13 suggests the PCIU is **NOT** definitive.
IGNORING EXAM: The PDF explicitly defines PCIU as a DEFINITIVE prosthesis.

Rarely Examined Areas / Future Targets

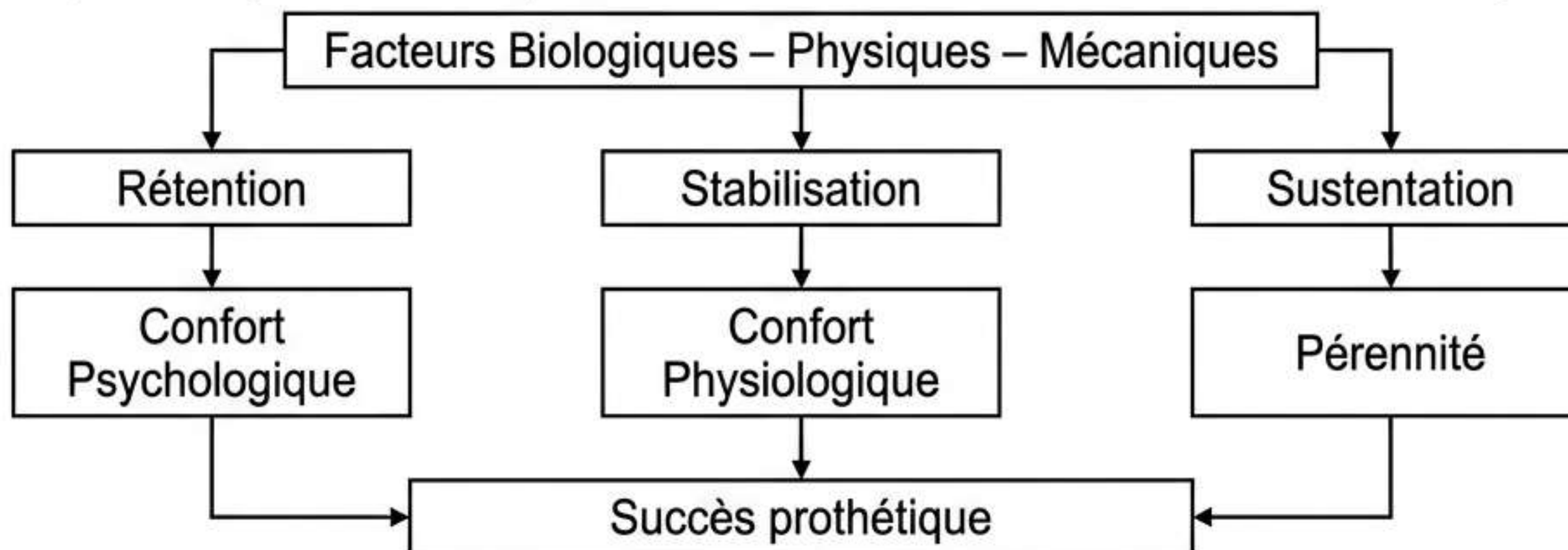
Material specificities (Polyether vs. Kerr), **specific medical contraindications** (Irradiation), and modern **Digital CAD-CAM workflows**.

Prediction Confidence: 92% based on recurring chronological questions.

The Mutilation of Edentulism & Housset's Triad (1925)

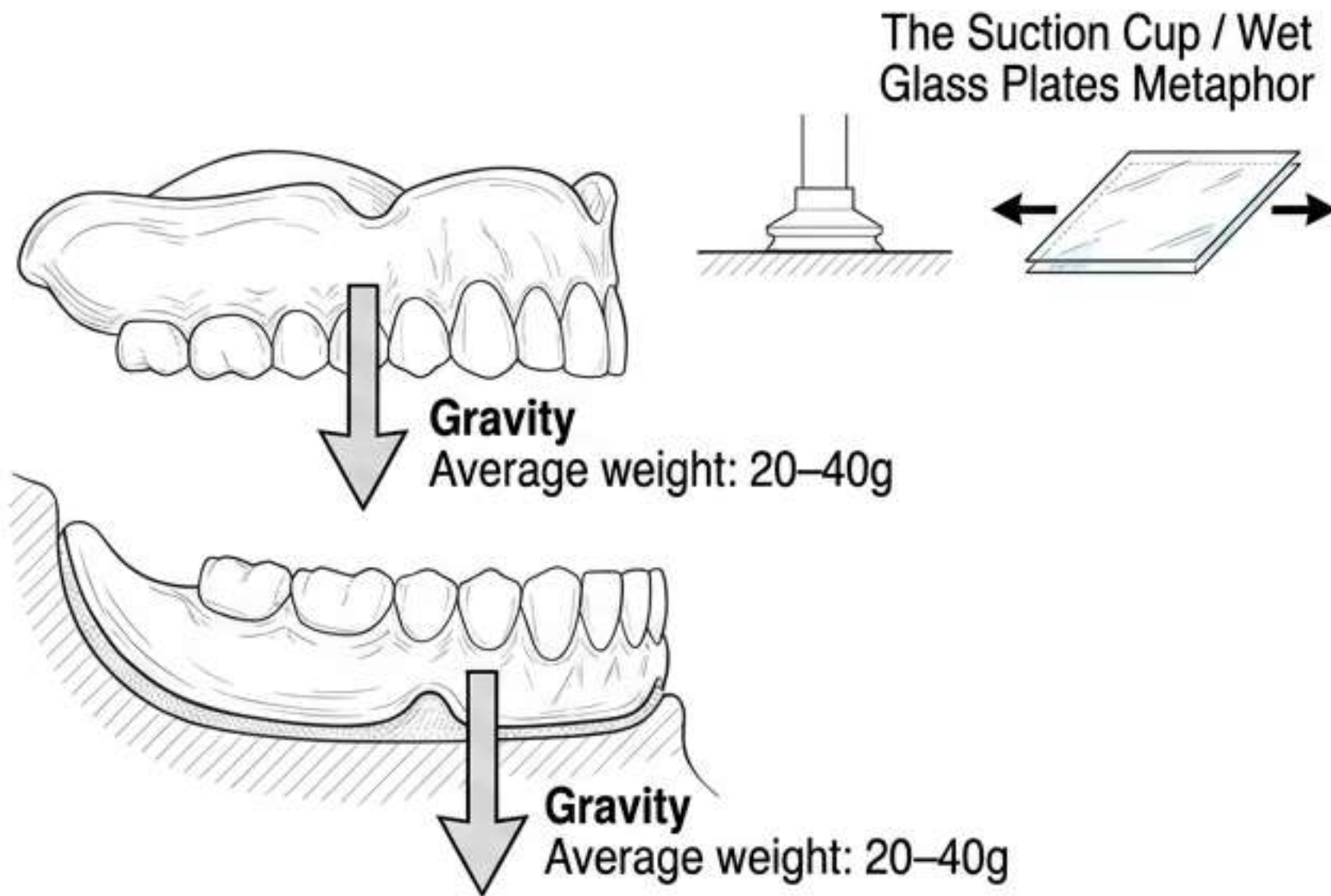
Total edentulism is a true **mutilation**. Prosthetic integration relies on exploiting support surfaces and perfect static/dynamic occlusal equilibrium.

The **Triad of Equilibrium** (Housset, 1925): Sustentation, Stabilization, and Retention are strictly interdependent.



Retention Definition	5 Roles of Retention
Favorable reaction opposing vertical forces that pull the prosthesis away from the supporting surface (Gravity at maxilla; bilateral traction of depressor muscles by sticky food).	1. Avoid disinsertion. 2. Assure functional efficiency. 3. Contribute to aesthetic credibility. 4. Favor psychological integration. 5. Preserve tissue integrity.

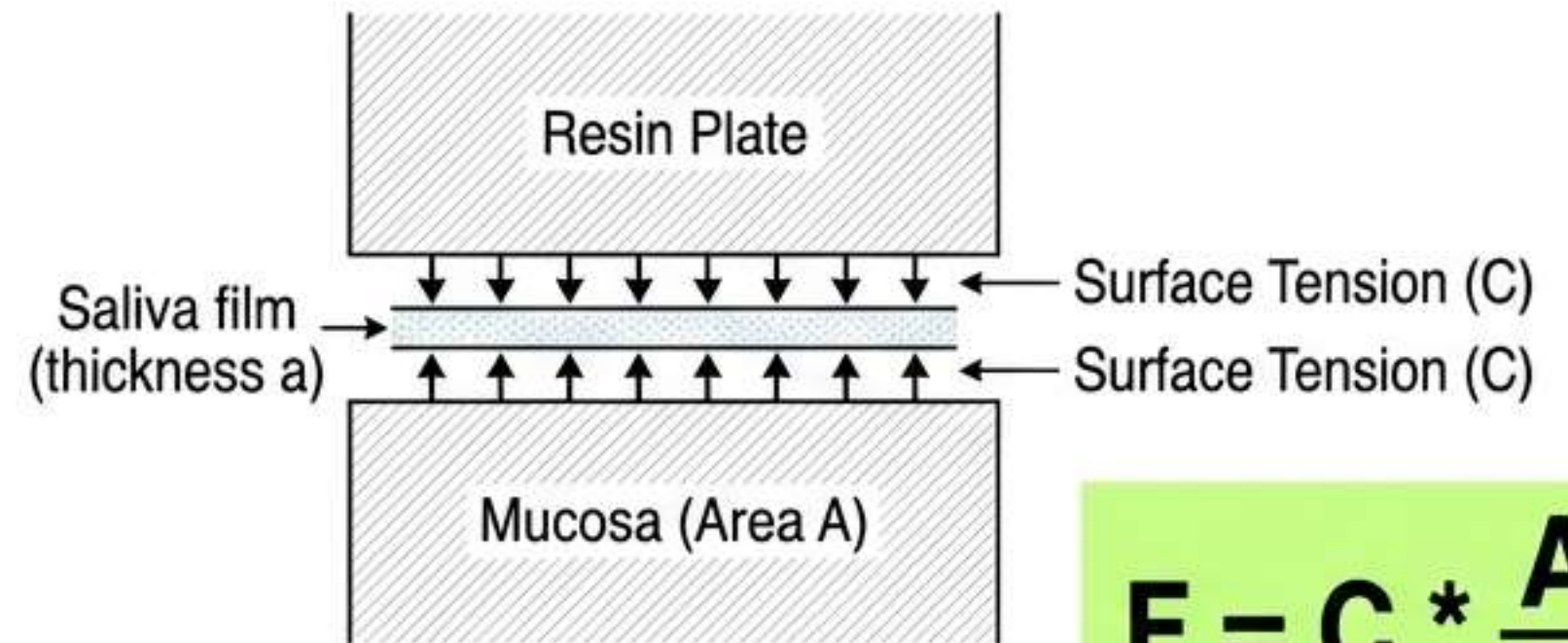
Retention Physics: Gravity, Atmosphere & The Staniz Formula



Gravity: Light at maxilla, heavy at mandible. Porcelain teeth/metal bases add weight.

Atmospheric Pressure:

- **Page:** Denies role
- **Boucher:** Emergency role only
- **Lejoyeux:** Active if peripheral seal exists; displacement creates relative vacuum.



$$F = C * \frac{A}{a}$$

Cohesion: Attraction between homogeneous molecules (saliva-saliva). Strongest with mucin-rich palatal saliva.

Adhesion: Attraction between different bodies (saliva-acrylic/mucosa).

Surface Tension & Wetting: High surface energy drives wetting. Resin coefficient < Metal.

Capillarity & Viscosity: Liquid meniscus resists separation. High viscosity (mucin) slows disinsertion speed.

Anatomo-Physiological & Neuromuscular Retention Factors

Fibromucosa & Contact Intimacy

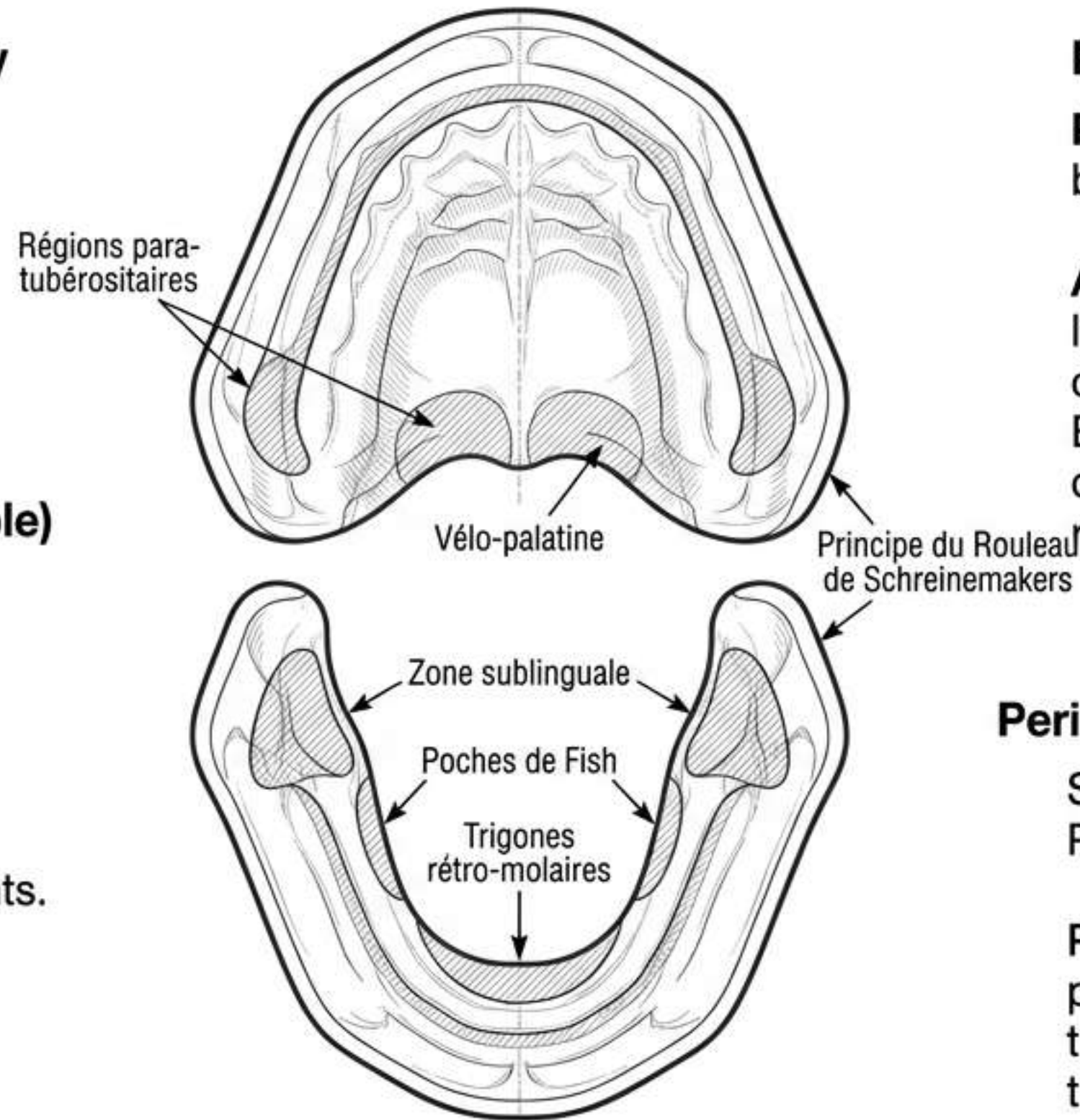
Requires rinsing, drying, custom trays, secondary functional impressions, and highly fluid impression material.

The Saliva Factor (Critical Variable)

Mandible has less retention due to higher salivary flow preventing an effective meniscus.

Decreased Secretion: Diabète, Syndrome de Gougerot-Sjögren, irradiations, ménopause, médicaments.

Increased Secretion: Grossesse, prothèses mal ajustées, adaptation period.



Neuromuscular Control

Passive: Weight of tongue, buccinator, orbicularis oris.

Active: Exteroceptors (jugal, labial, lingual) transmit to trigeminal nucleus causing reflex muscle reactions. Brill's study: Contact anesthesia disorganizes stereognosis, reducing retention.

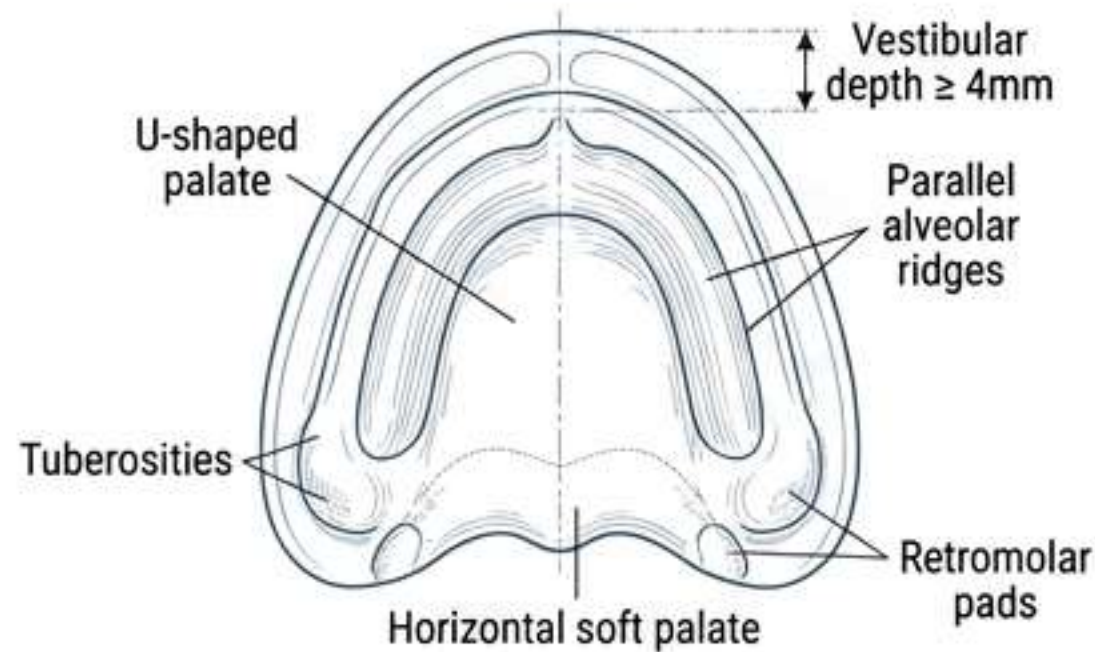
Peripheral Seal & Surface Extension

Surface Extension maximizes Positive Indices.

Peripheral Seal opposes air penetration. Follows the Principle of the Schreinemakers Roll (smooth, thick, rounded borders on compressible tissue).

Mechanical, Psychological & Surgical Intervention

Mechanical Mastery

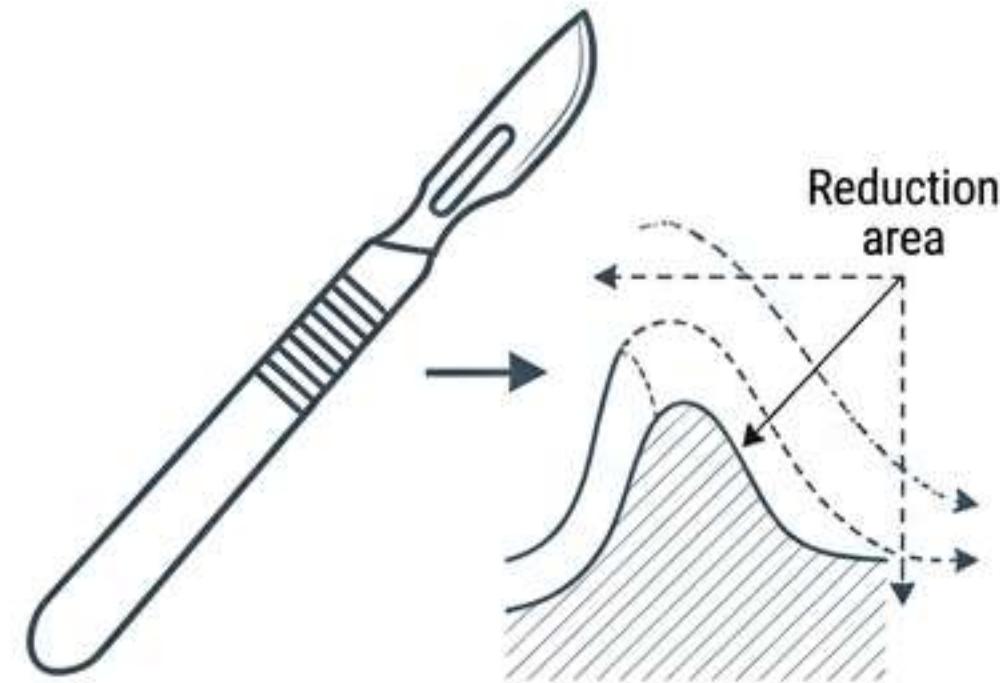


Favorable Anatomy: Dense/adherent fibromucosa, vestibular depth $\geq 4\text{mm}$, wide/high parallel ridges, U-shaped palate, good tuberosities, horizontal soft palate, hard marked retromolar pads, internal oblique line (LOI) not too prominent.

Phonetic & Aesthetic: Lingual support must allow pronunciation without mechanical imbalance.

Occlusal: Entirely **balanced occlusion** guarantees retention; errors cause instability and rapid bone resorption.

Surgical Factors



Tissue/Bone Surgery: Resection of mucosal bands (brides), flabby ridges (crêtes flottantes).

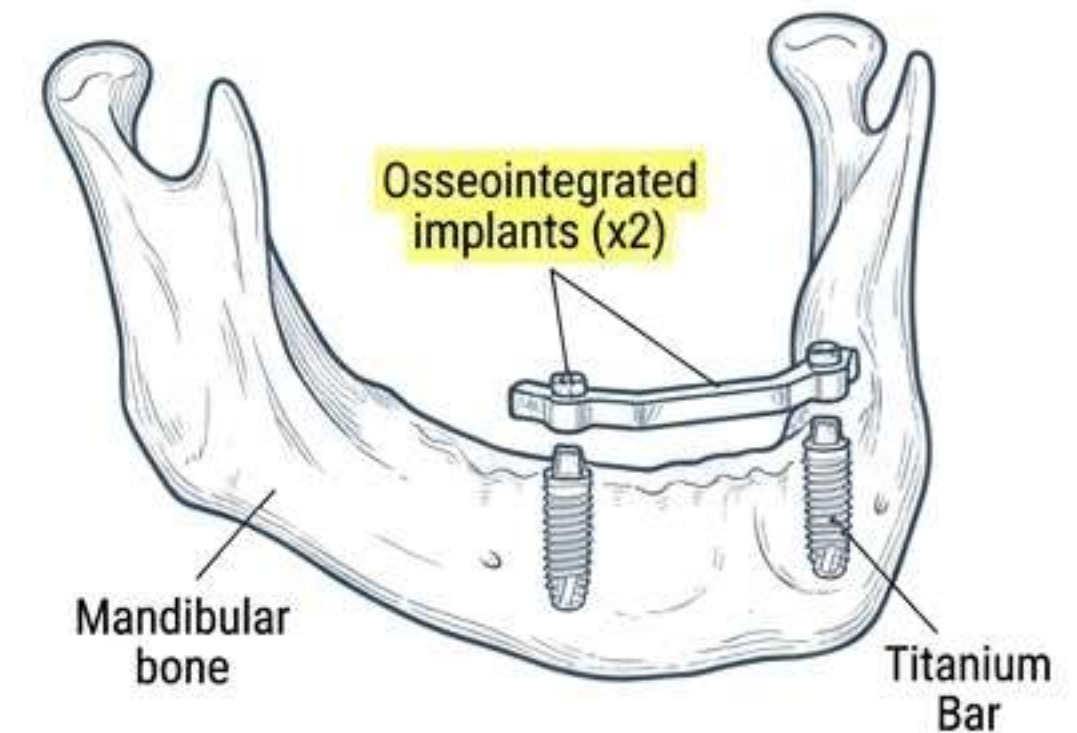
Lowering floor of mouth (**Trauner operation**).

Vestibular deepening (with/without graft).

Bone grafts.

Remodeling tori/tuberosities/LOI.

Implantology & Adhesives

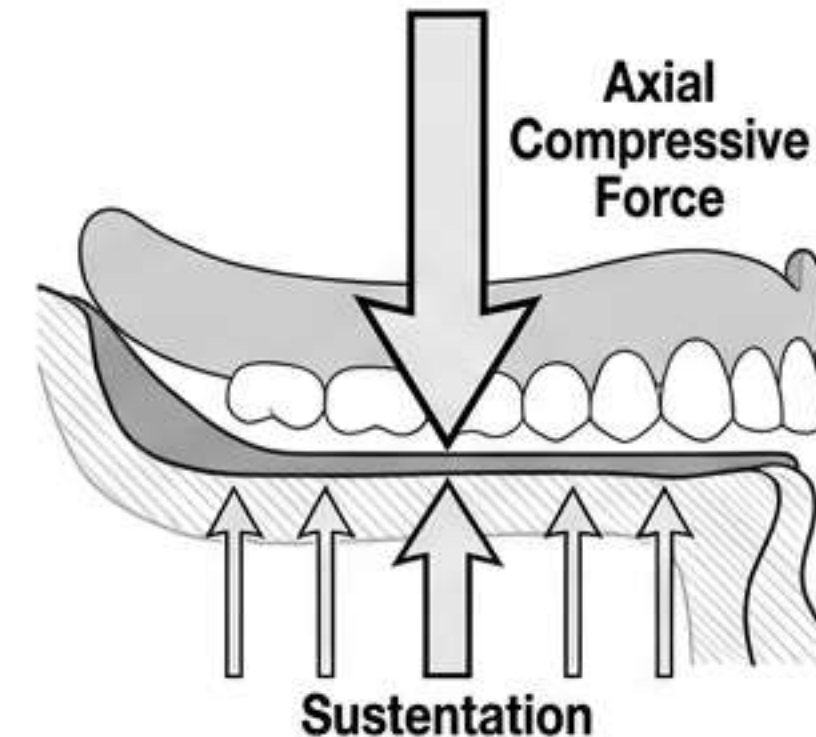
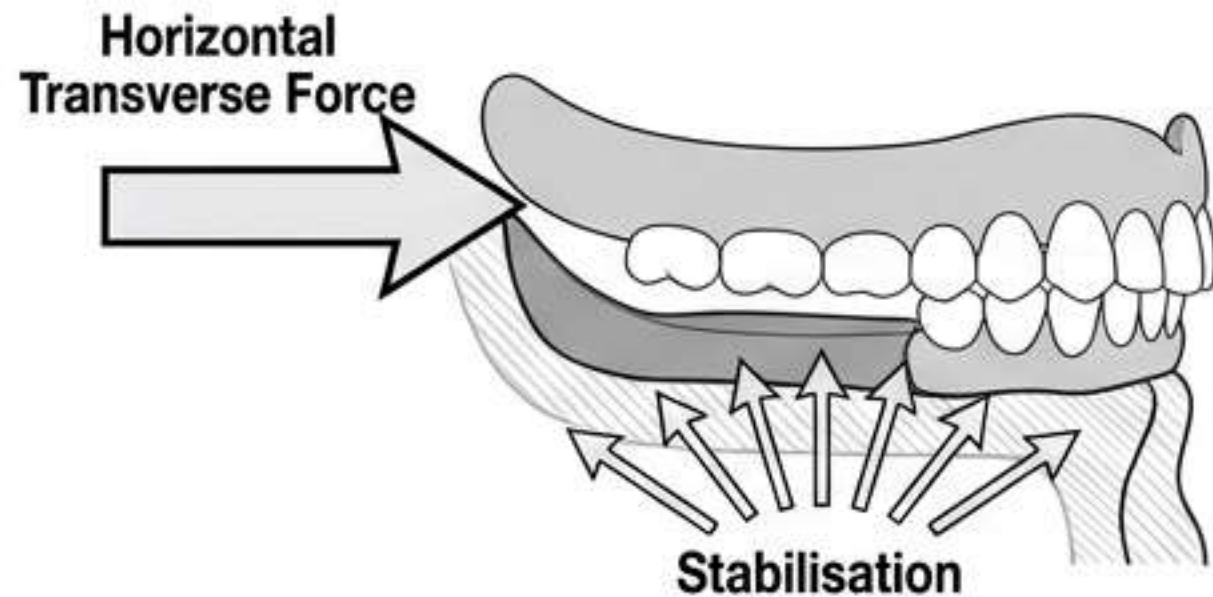


Implantology: 2 implants at mandible, 4 at maxilla.

Retention systems: Barres, bouton-pression (buttons), **aimants** (magnets).

Chemical Adhesives: Increase salivary viscosity to retard disinsertion. Brands: Corega, Poli-Grip, Fixodent, Stéradent.

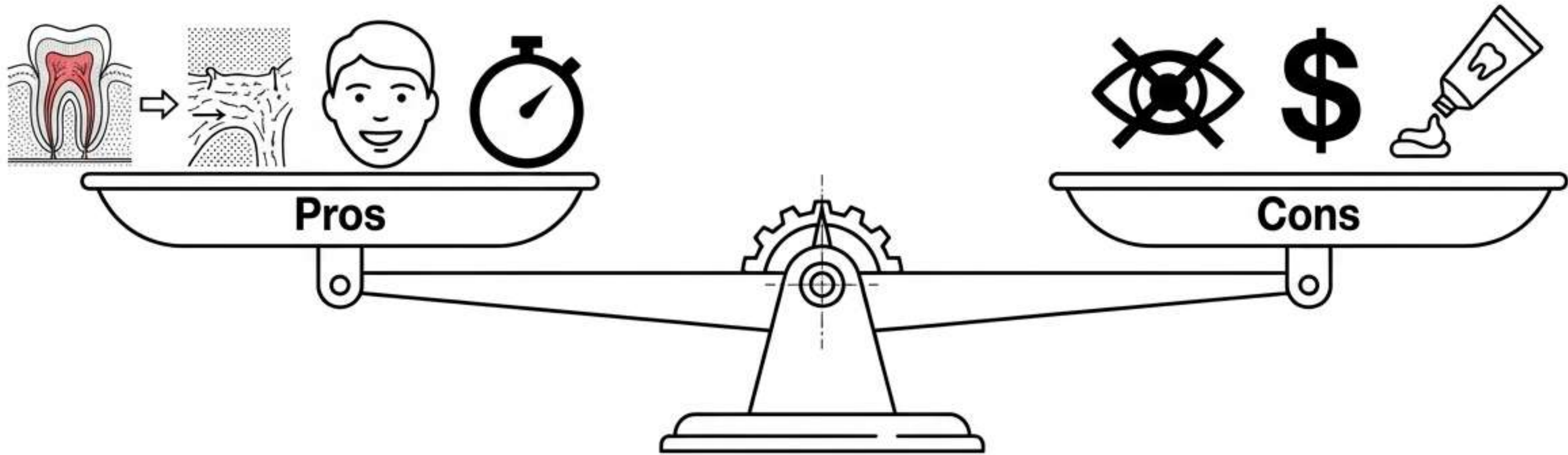
Stability (Horizontal) & Sustentation (Axial)



STABILITY (Resistance to Transverse/Rotation Forces)	SUSTENTATION (Resistance to Axial Sinking)
Definition (Batarec): Reaction opposing translation or rotation under muscular forces. FISH's Criteria: • I: Bases must not interfere with muscle attachments. • II: Neutralize horizontal forces by buccinator-labial sling and tongue. • III: Cancel destabilizing forces from occlusion. (Only inter-dental contacts can dislodge the prosthesis). Anatomical Helpers: • Maxilla: Wide/parallel ridges > flat/triangular; Square arch > ovoid; Flat vault > ovoid. • Mandible: Retromylohyoid regions are absolutely critical due to negative ridges.	Definition: Resistance of osteo-mucosal tissues to the sinking of the prosthesis. Factors of Maximization: • Maximizing surface area (Force/Area ratio). • Favorable U-shaped palate with a wide horizontal base. • Ideal wide/high parallel ridges. • Tuberosities parallel to occlusal plane. • Mandibular cortical bone must be smooth without spines. • All negative indices must be relieved/bypassed.

Immediate Complete Dentures (PCIU): Definition & Biological Balance

Definition (Lejoyeux): A **DEFINITIVE** prosthesis designed and fully realized **BEFORE** extraction, and inserted **IMMEDIATELY** after [Ref: Q16]. **Objective:** Maintain last maxillary anterior teeth during elaboration.



Advantages (Heavy Exam Focus)

- Acts as a **TRUE SURGICAL BANDAGE** (Pansement chirurgical) [Ref: Q11, Q13]
- **Protects the wound and organizes the blood clot** [Ref: Q5, Q15]
- **Decreases (limits) post-extraction bone resorption** [Ref: Q19, Q84, Q93]
- **Avoids the transitoire edentulism period** [Ref: Q7]
- **Preserves aesthetics and vertical dimension (DV)** [Ref: Q15]
- **Maintains neuromuscular equilibrium and phonetics.**

Disadvantages & Realities

- **Frequent relining (rebasage) is absolutely necessary** [Ref: Q12, Q13, Q18]
- **No possibility of an anterior aesthetic try-in** [Ref: Q18]
- **High cost and complex post-operative care.**

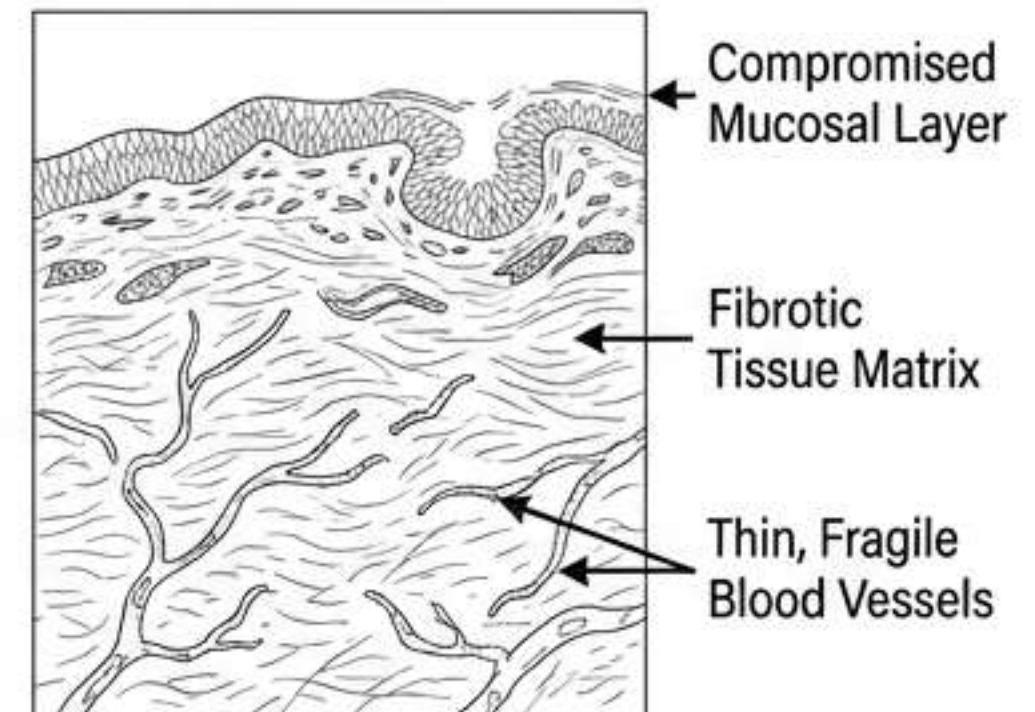
Clinical Diagnostics: Indications & Contra-indications

INDICATIONS (The Green Light)

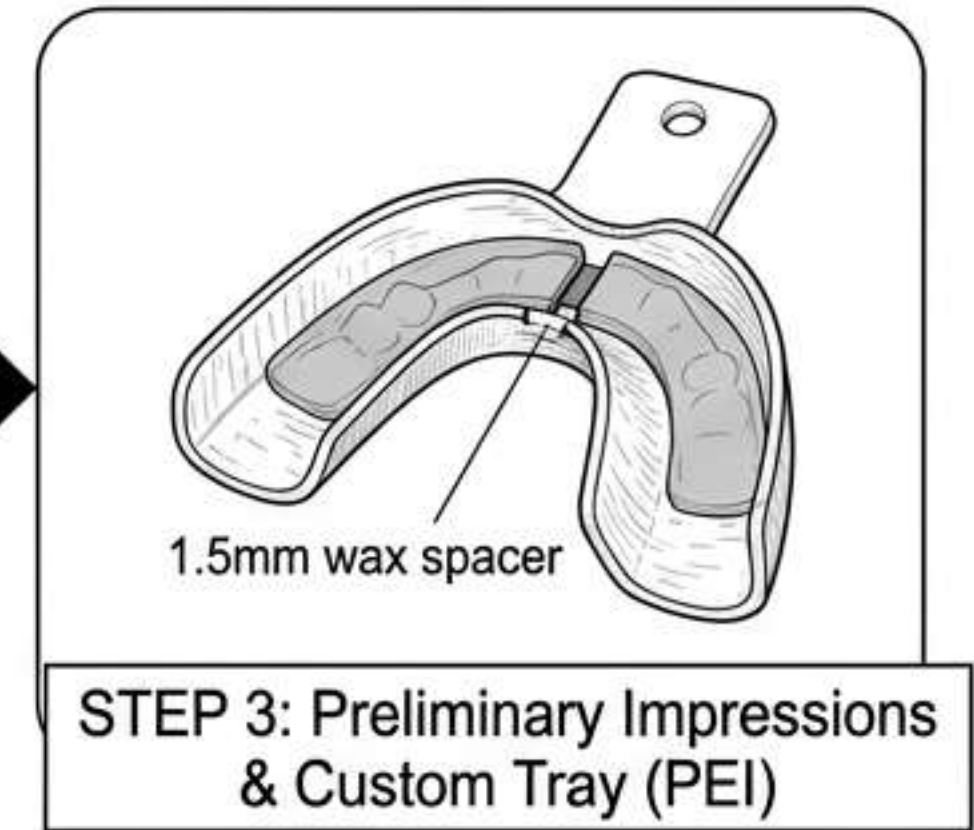
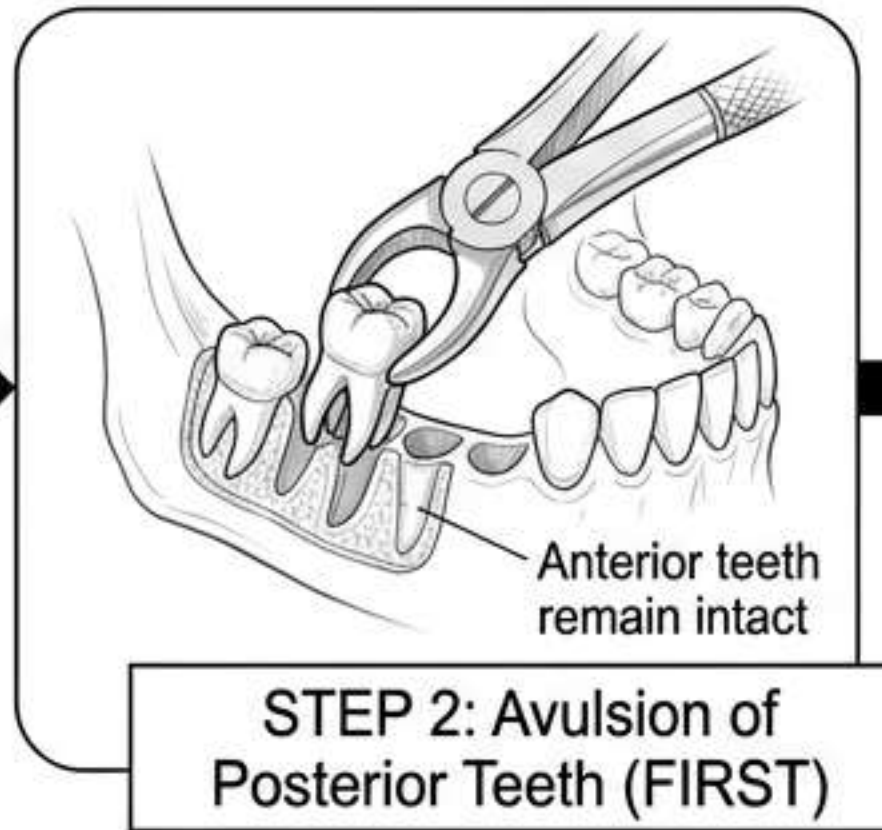
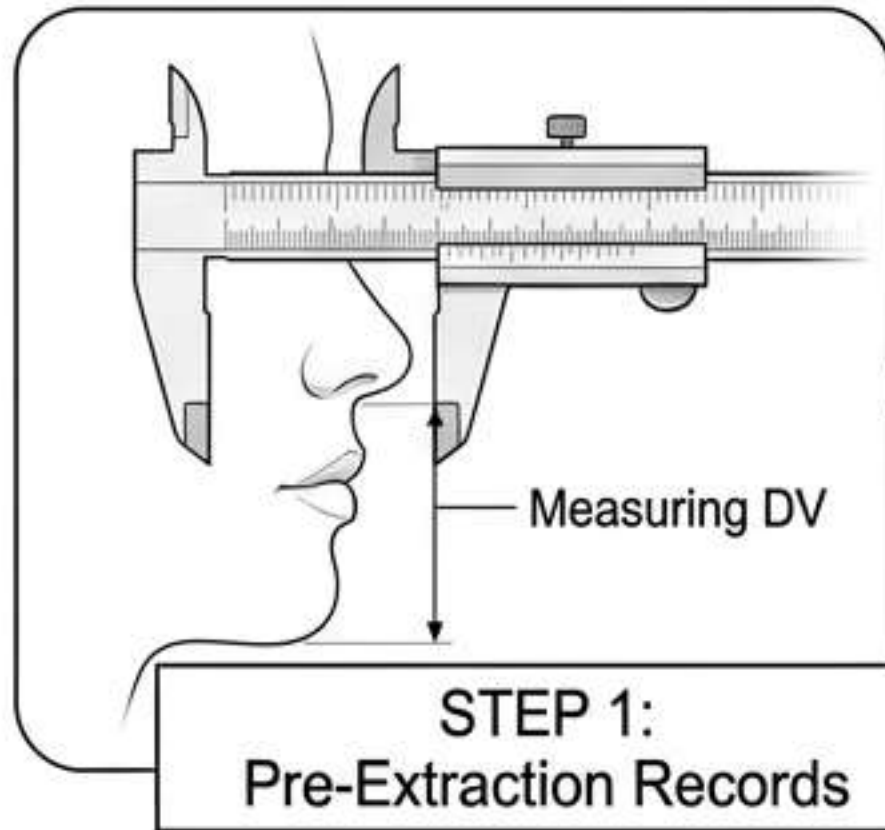
- Anxious patients terrified of appearing edentulous.
- Public-facing professions (journalists, artists).
- Urgent prosthetic treatment needs (travel, vacation, marriage).
- General grave affections where an infectious focus is a vital risk (e.g., Cardiopathy with high infectious risk).
- Insufficient intrinsic/extrinsic value of remaining teeth.

CONTRA-INDICATIONS (The Red Light)

- Patients with medical states presenting surgical risks (blood disorders, severe cardiac conditions, uncontrolled diabetes).
- Irradiated Patients: Compression of fragile, poorly vascularized tissue can have disastrous consequences [Predicted - Biological trap].
- Unmotivated or non-cooperative patients (post-therapeutic follow-up is impossible).



Clinical Workflow Part I: Records & The Posterior First Rule



Step 1: Pre-Extraction Records

- Record Vertical Dimension (DV) – The most precious pre-extraction element.
- Record Occlusal Plane (balanced with tongue/buccinator).
- Record shade, shape, dimensions, and position of natural anterior teeth. Pre-op cast taken.

Step 2: Avulsion of Posterior Teeth (FIRST)

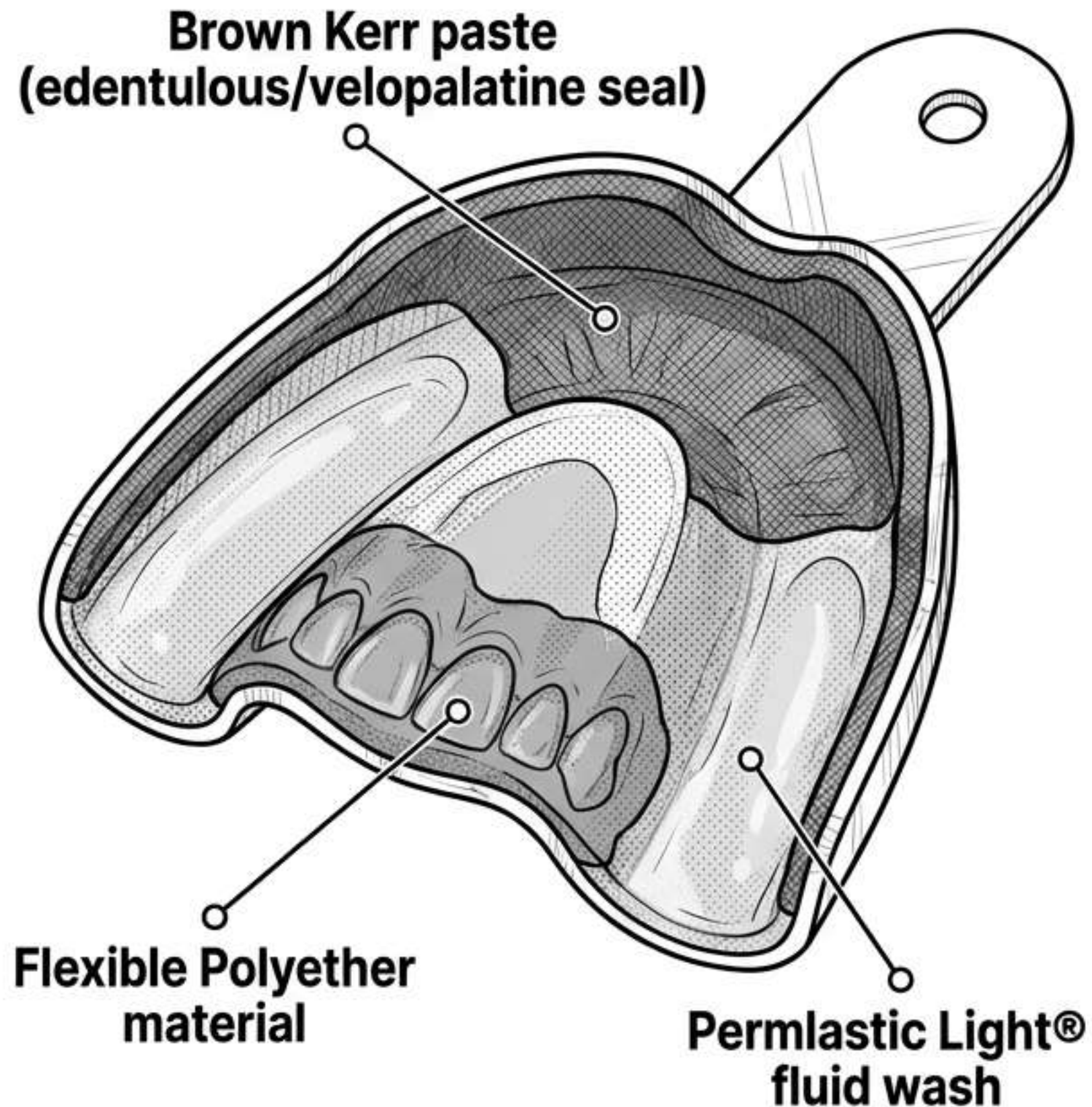
Avulsion of Posterior Teeth (FIRST) [Ref: Q14, Q20]

- Posterior teeth are extracted first. Must be minimally invasive to preserve bone.
- Wait Period: 3 to 4 weeks of healing before initiating the prosthetic treatment.

Step 3: Preliminary Impressions & Custom Tray (PEI)

- Preliminary impression records remaining anterior teeth and edentulous posterior sectors.
- **Laboratory:** Remaining plaster teeth are covered with a 1.5mm calibrated wax sheet to create a future spacer. The resin tray (PEI) adapts to the mucosa but is spaced over the teeth.

Clinical Workflow Part II: Dual-Material Impressions & Try-In



Step 4: Secondary Functional Impressions

- **Peripheral/Posterior Seal:** Recorded with Kerr paste in edentulous and velopalatine regions.
- **Anterior Teeth Region:** Recorded with a flexible material (Polyether) to capture remaining teeth [Predicted - Material knowledge].
- **Wash:** Full impression taken with Permlastic Light®.

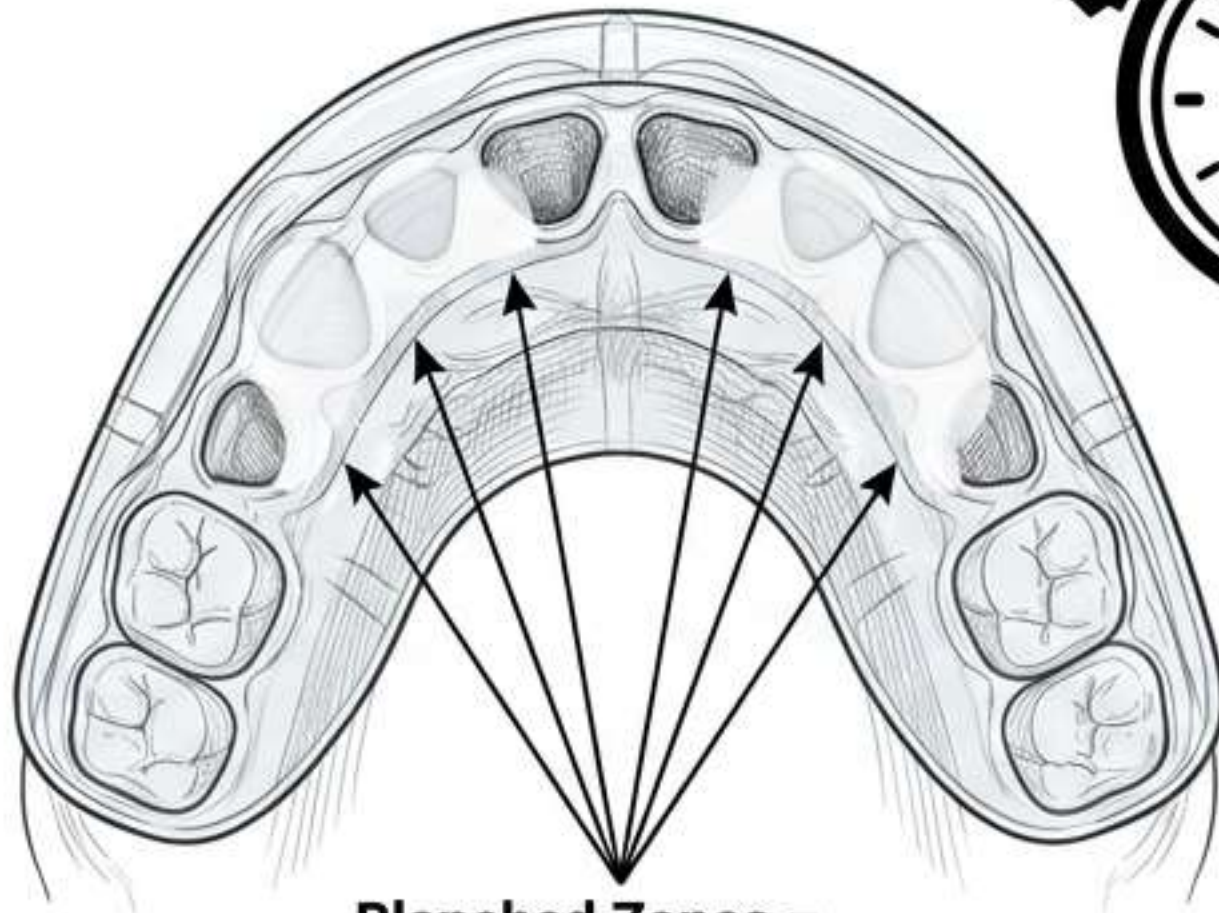
Step 5: Lab Treatment & Intermaxillary Records (RIM)

- Cast poured, occlusion rims made.
- Determine DVR, DVO, and Centric Relation (RC). Transfer to articulator.

Step 6: Setup & Functional Try-In

- Posteriors (premolars/molars) chosen and mounted in fully balanced occlusion.
- **Try-in:** Validate aesthetics, DVO, and intercuspation.
- **Crucial Clinical Step:** Practitioner measures periodontal pockets (vestibular, mesial, distal) of remaining teeth and marks exact depths in pencil on the cast.

Surgery, Immediate Insertion & The 48-Hour Protocol



**Blanched Zones =
Tissue Compression
(Requires Bone Plasty)**

1

Surgical Act & Insertion

- **Vestibular** flap raised, anterior teeth extracted.
- **Bone Plasties:** Controlled by inserting the transparent surgical guide. Blanching indicates compression and signals exact zones to reduce. Correct until uniform pressure is achieved.
- **Suture** vestibular to palatal flap with strictly two suture points.
- **Immediate Insertion:** Inserted instantly. Patient bites on cotton rolls for several minutes to distribute edema and stop hemorrhage. Occlusion checked (only obvious posterior contacts corrected; no anterior contacts).

2

Post-Operative Instructions (Strict Rules)

- **Do NOT remove prosthesis for 48 hours** (edema will prevent reinsertion).
- Apply local ice. Take analgesics. Chlorhexidine mouthwash.
- **Liquid/soft diet only** [Ref: Q7]

3

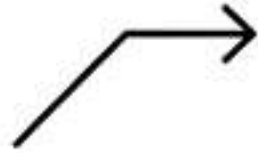
Maintenance & Control

- **At 48h:** Practitioner removes, cleans, and relieves sore spots.
- Wear 24/7 for the first week to protect clot.
- **Mandatory follow-ups:** 1 week, 3 months, 6 months, 1 year (relining inevitable).

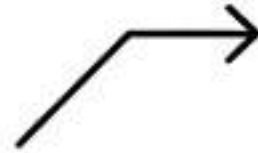
Provisional Variants & The Digital Workflow (CAD-CAM)



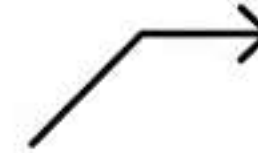
Intraoral Scanner



3D Rendering



Virtual Extraction



CAD-CAM Milling

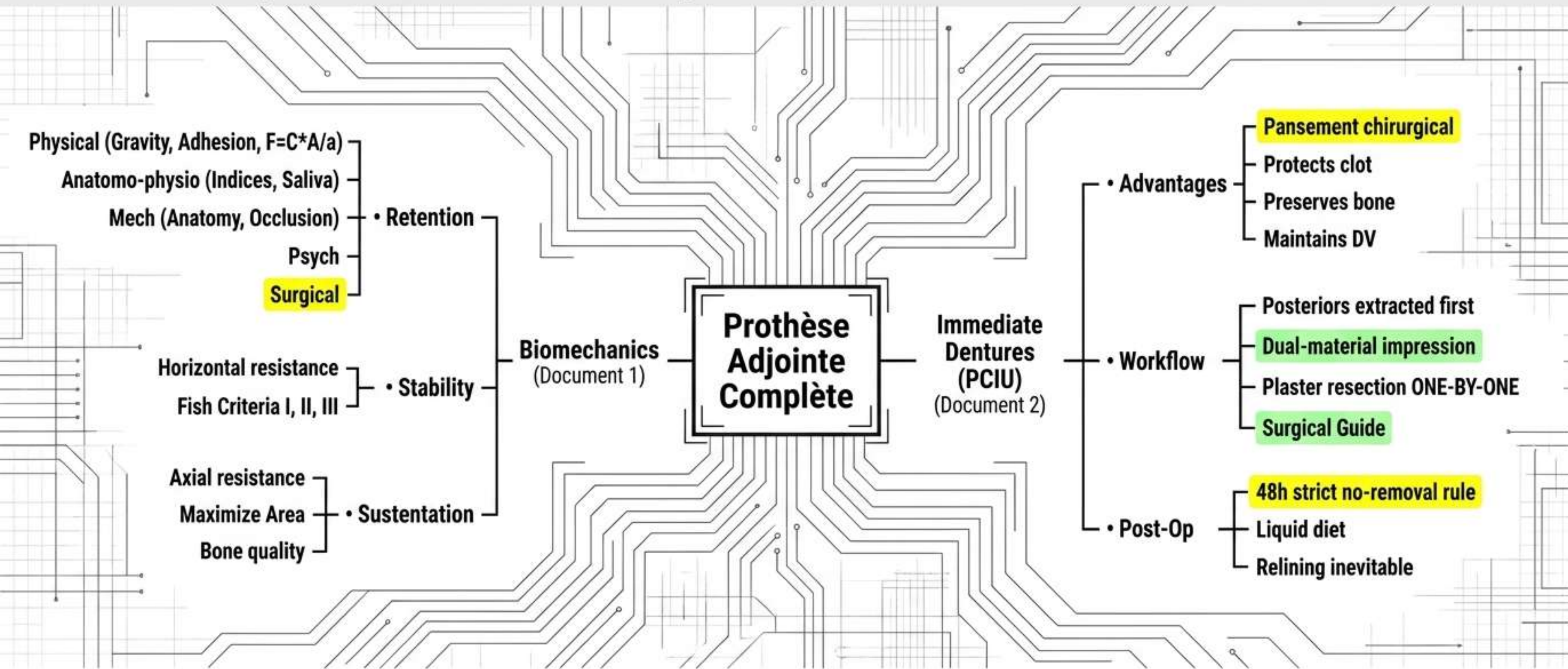
The Provisional PCIU Variant

- Simplified protocol.
- Created using only primary impressions (no secondary anatomic-functional impression).
- Used temporarily during the healing phase (3 to 6 months) before being replaced by a definitive prosthesis.

Apport du Numérique (The Digital CAD-CAM Era)

- Digital Prediction: **Intraoral scanners** and **3D software** allow for **highly precise Virtual Extractions** [Predicted - Tech evolution].
 - Anticipates hard and soft tissue modifications computationally.
 - **Optimizes** prosthetic tooth positioning with zero physical plaster resection.
 - **Guides** and prostheses are milled/printed via **CAD-CAM**, reducing post-operative adjustments and improving predictability.

Complete Macroscopic Synthesis & Certification



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